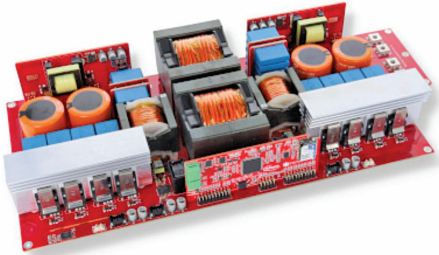


Interesting Reference Designs Of

Title of Reference Design	Bi-Directional Converter for EV Charging	6.6kW Unidirectional On-Board Charger
Image		
Key Attractions	The reference design is for an 11kW SiC bi-directional DC/DC converter board. It's intended for electric vehicle (EV) charging and energy storage system (ESS) applications. This design offers a solution for bi-directional DC-DC converters. It enables customers to create rapid and unique bi-directional charger designs.	The reference design is for a 6.6kW unidirectional on-board charger (OBC). It showcases the capability of 22mΩ power GaN FET transistors. These transistors enhance OBC efficiency. They also boost power density by operating at higher switching frequencies.
Highlights	The REF-DAB11KIZSICSYS, a reference design from Infineon Technologies, is an 11kW SiC bi-directional DC/DC converter for EV and ESS applications. Its CLLC resonant converter supports bi-directional energy flow and soft-switching, which is ideal for onboard and off-board chargers and ESS. The design ensures smooth integration of CoolSiC MOSFETs with Infineon's suite of components, including the driver IC, XMC controller, flyback controller, voltage regulator MOSFETs, current sensor, Cypress memory, and a security & safety chip. The central feature is the CLLC in a full-bridge configuration, chosen for its role as a DC transformer linking the AC/DC to the DC bus. In both forward and reverse operations, its resonant tank reflects the characteristics of a traditional LLC resonant tank, achieving soft switching with zero voltage switching (ZVS) and zero current switching (ZCS) in both directions. This capability cuts down on switching losses and boosts charger efficiency. Remarkably, the design consistently maintains an efficiency of 97.2% or more. Boasting a power density of 4.1kW/L, it integrates synchronous rectification and features both 1200V and 1700V MOSFETs. It supports cutting-edge business models like vehicle-to-grid (V2G) and vehicle-to-building (V2B), ensuring minimal losses during any charge or discharge cycle.	The VRD-UNIOBCV1M1, a reference design from VisIC Technologies, incorporates six gallium nitride (GaN) devices with two in the dual boost bridgeless power factor correction (PFC) and four in the full-bridge LLC (inductor-inductor-capacitor) with diode rectification. GaN benefits enable smaller PFC inductors and LLC transformers, ensuring lower power losses with fewer devices. Additionally, it showcases efficient heat dissipation using isolated top cooling. The OBC boasts several features, including a high power density of 3kW/L and an efficiency rate of 96%. The charger can accommodate an input of 90V-264V/47-63Hz in a single phase and provides an output ranging from 240V to 420V. It can deliver power up to 6.6kW with a 20A current at its maximum capacity, using liquid cooling through a cold plate. For safety and functionality, the charger includes protections such as input under-voltage protection (UVP), over-voltage protection (OVP), and over-current protection (OCP). Remarkably, it has a power factor (PF) of 0.99 at full load and offers load constant current (CC), constant voltage (CV), and constant resistance (CR) modes.
Applications	The reference design is optimal for both on-board and off-board chargers and energy storage systems (ESS).	The reference design is suitable for a quick charging solution without overburdening residential and public electrical systems.
OEM Brand	Infineon Technologies	VisIC Technologies
URL	https://www.electronicsforu.com/electronics-projects/reference-design-of-a-bi-directional-converter-for-ev-charging	https://www.electronicsforu.com/electronics-projects/reference-design-for-6-6kw-unidirectional-on-board-charger